Thermodynamic limits of kinetic-only separation

A finite-contrast separation potential and constructive common controls

Research continuation

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The result and its boundary

The independently important question is whether chemical species with the same equilibrium interactions can be separated efficiently using only differences in their rates of motion or reaction. Programmable ratchets and ordinary countercurrent separation already provide extensive sorting capabilities. The new object developed here is a thermodynamic state function that charges the complete sorting task: useful product, recovery, rejects, and retained internal information.

For a declared common-affinity stochastic dynamics, a finite-contrast convex potential gives a lower bound on *actual integrated entropy production* under arbitrary time dependence, intermediate storage, and numbers of stages. The proof covers both controlled diffusions and finite-jump networks. A shared moving harmonic potential gives an exactly solvable finite-time construction, including all driving work and residual free energy. Its weak separation limit has precisely the local cost required by the potential. An exact two-state cell then gives a physical retained-history construction approaching the global bound in an ideal common-control limit. Routing and reset are explicitly included in the construction; finite apparatus losses remain a separate engineering task.

The local efficiency function is classical Kedem–Caplan coupling algebra. The weak-contrast potential is the established Dirac separation value. Neither is claimed as newly discovered. The specific contribution constructed here is their finite-contrast, nonstationary endpoint connection, its exact network extension, and a matching physical construction of the ideal infimum. The selective primary-source search did not locate the same full theorem; historical priority is not established. No separator apparatus, measured material response, or practical advantage has been demonstrated.

The preceding rare-risk, source-likelihood, linkage, preparation, port-lifetime and complete-excursion research is preserved unchanged and parked.

1 Why the first proposed mechanism was rejected

The initial intended use was valuable-ion recovery without highly selective binding materials. The personally constructed candidate used multiple traveling electric waves in a self-consistent electrolyte. Linearized Poisson–Nernst–Planck response yields, for species i,

$$v_i = \frac{\beta^2 q_i^2 |\phi|^2 \omega}{2k_x} \frac{D_i^2}{D_i^2 + (\omega/k_x^2)^2}, \qquad \phi = \frac{\phi_{\text{ext}}}{1 + \frac{\beta}{\varepsilon k_x^2} \sum_j q_j^2 n_j \frac{D_j k_x^2}{D_j k_x^2 - i\omega}}.$$

Distinct wave frequencies give a Cauchy response matrix for distinct diffusivities. This suggested independently programming species currents. However, Jansons and Lythe already derived

the multiwave rational diffusivity kernel in 1998, and Bogunovic et al. already constructed programmable multispecies transport in 2012 [1, 2]. Adding screening and a local continuation argument would not establish a new broad sorting principle.

An independent timed-gate construction attained high purity and recovery, but its exact absorption law was reproduced by ordinary countercurrent drift and diffusion. A neutral-flow constructor found a common-invariant-density obstruction and a passive-chamber recovery limit, rather than a new useful separator. Those derivations and failures remain executable in the archive. The research question therefore changed from constructing another sorter to identifying the thermodynamic limit of *all* common-energy kinetic sorters.

2 The physical model and what is being counted

Let S be a conserved species label, with prior probabilities $w_s > 0$. The conditional position densities $p_s(x,t)$ are normalized. The per-particle interpretation is an independent-tracer or ideal-mixture model. Equivalently, the network theorem concerns a single conserved binary label and the full controlled state associated with it. Extending its sharper binary cost to a sum over many interacting unknown labels requires an additional theorem; no such extensive collective claim is made here. Set

$$m = \sum_s w_s p_s, \qquad r_s = \frac{w_s p_s}{m}.$$

The species experience the same force $F(x,t)$ and common transport velocity $v(x,t)$, but have different positive diffusivities D_s. With $\beta = (k_B T)^{-1}$, write

$$j_s = v p_s - D_s p_s u_s, \qquad u_s = \nabla \log p_s - \beta F, \qquad \partial_t p_s = -\nabla \cdot j_s.$$

Use closed periodic/no-flux accounting, or include every collector and reject reservoir explicitly. The dimensionless diffusive entropy-production rate is

$$\sigma = \sum_s w_s \int D_s p_s |u_s|^2 \, dx, \qquad \Sigma = \int_0^T \sigma \, dt. \tag{1}$$

For $v = 0$, this is the usual isothermal diffusion entropy production. If v denotes common solvent or material advection, (1) is the relative-diffusion contribution; solvent-flow dissipation is additional. It must not be identified with the laboratory position-path expression $\sum_s w_s \int |j_s|^2/(D_s p_s)$ for an arbitrary imposed drift. All upper constructions below can be stated without this advection extension. Physical entropy is $k_B \Sigma$. Actuators and controllers can add dissipation; their costs cannot be subtracted from (1).

The restriction is *shared thermodynamic forces*, not merely use of one voltage wire. Different valences, binding energies, polarizabilities, accessible paths, or chemical affinities may violate it. An effective Brownian model must be warranted for the actual operating range. A numerical bulk diffusion ratio does not establish binding/release kinetics in a material.

The finite-state counterpart permits arbitrary time-dependent graphs with forward/backward rates

$$k_{xy}^{(s)} = a_{xy}^{(s)} k_{xy}, \qquad k_{yx}^{(s)} = a_{xy}^{(s)} k_{yx}.$$

The same species multiplier acts in both directions of each edge. Thus the edge affinity $\log(k_{xy}/k_{yx})$ is common. The multiplier can vary by edge and time, subject to a declared finite contrast. Edges may be closed for all species together. Species-specific forbidden edges are a different resource. The state space can include intermediate pools, routing records and controllers; their actual transitions must obey the same restrictions for the theorem to cover them.

3 An information bound with an explicit kinetic contrast

Define instantaneous stored species information

$$\mathcal{I}(S;X) = \sum_s w_s \int p_s \log(p_s/m)\, dx.$$

Common advection contributes zero to its derivative after full boundary accounting. Integration by parts gives

$$\dot{\mathcal{I}} = \int m\{\langle Du\rangle_r \cdot \langle u\rangle_r - \langle D|u|^2\rangle_r\}\, dx. \tag{2}$$

Theorem 1 (Stored information and entropy production)**.** *If $0 < a \le D_s \le b$, then*

$$\mathcal{I}_T - \mathcal{I}_0 \le \alpha\Sigma, \qquad \alpha = \frac{(\sqrt{b}-\sqrt{a})^2}{4\sqrt{ab}} = \sinh^2\!\left(\frac{1}{4}\log\frac{b}{a}\right).$$

The same conclusion holds for finite-jump networks when the species multiplier ratios on every edge have the corresponding uniform bound.

Proof. At a fixed position and for one vector component put $z_s = \sqrt{r_s D_s}u_s$, $A_s = \sqrt{r_s D_s}$ and $B_s = \sqrt{r_s/D_s}$. The symmetric rank-two matrix $(AB^T + BA^T)/2$ has largest eigenvalue

$$\frac{A\cdot B + \|A\|\|B\|}{2} = \frac{1+\sqrt{\langle D\rangle\langle 1/D\rangle}}{2}.$$

Consequently the integrand of (2) is at most $\alpha(r)$ times that of (1), where $\alpha(r) = (\sqrt{\langle D\rangle\langle 1/D\rangle} - 1)/2$. The pointwise inequality $D + ab/D \le a + b$ and AM–GM give $\langle D\rangle\langle 1/D\rangle \le (a+b)^2/(4ab)$. Integrating proves the diffusion result. The edge interpolation in the next section gives the exact jump extension; it is not a diffusion approximation. □

For $b/a = 1+\delta$ with small δ, $\alpha \sim \delta^2/16$. Thus weak kinetic contrast imposes an inverse-square entropy penalty for creating a fixed amount of stored label information. Extra duration alone does not remove that penalty within this model.

For an isothermal common conservative energy, write the joint nonequilibrium free energy as $F_{\text{joint}} = F_{\text{blind}} + k_B T\mathcal{I}$. The full work balance is

$$\beta W = \beta\Delta F_{\text{blind}} + \Delta\mathcal{I} + \Sigma.$$

Only when the species-blind free energy and other machine resources are restored does this imply $\beta W \ge (1 + 1/\alpha)\Delta\mathcal{I}$. Initially prepared packets, empty memories and chemical fuel can supply free energy and must be charged.

4 The stronger finite-contrast separation potential

For two species write $D_1 = a, D_2 = b$, prior f, posterior $r = fp_1/m$, and

$$t_r = r(1-r), \quad K = \frac{(a-b)^2}{ab}, \quad A = ar + b(1-r), \quad Q = a(1-r) + br.$$

Define a convex function ψ by

$$\boxed{\psi''(r) = \frac{2[1+\sqrt{1+Kr(1-r)}]}{Kr^2(1-r)^2}, \qquad \psi(f) = \psi'(f) = 0.} \tag{3}$$

Affine normalization has no effect on its endpoint differences. Put $\mathcal{F}_\psi = \int m\psi(r)\, dx$.

Theorem 2 (Complete endpoint cost)**.** *For the two-species common-force model,*

$$\boxed{\Sigma \geq \mathcal{F}_\psi(T) - \mathcal{F}_\psi(0).} \tag{4}$$

Starting from a mixed state, any final outlet variable Y therefore obeys

$$\Sigma \geq \sum_y P(Y=y)\psi(P(S=1 \mid Y=y)).$$

The theorem holds exactly for arbitrary finite-jump networks with common edge affinities. If contrasts vary, use K at least as large as $(a_e - b_e)^2/(a_e b_e)$ on every active edge and time.

Diffusion proof. Let $h = \nabla \log m - \beta F$ and $g = \nabla r$. Direct differentiation gives

$$\begin{aligned} \sigma_x/m &= A|h|^2 + 2(a-b)h \cdot g + (Q/t_r)|g|^2, \\ \dot{\mathcal{F}}_{\psi,x}/m &= -\psi''(r)\{Q|g|^2 + (a-b)t_r h \cdot g\}. \end{aligned}$$

For a proposed curvature $c \geq 0$, the quadratic form $(\sigma_x - \dot{\mathcal{F}}_{\psi,x})/m$ has matrix

$$\begin{pmatrix} A & (a-b)(1+ct_r/2) \\ (a-b)(1+ct_r/2) & Q(1/t_r + c) \end{pmatrix}.$$

Using $AQ = ab + (a-b)^2 t_r$, its determinant is

$$\frac{ab}{t_r} + abc - \frac{(a-b)^2 c^2 t_r^2}{4}.$$

The largest nonnegative curvature making this matrix positive semidefinite is exactly (3). Hence $\dot{\mathcal{F}}_\psi \leq \sigma$. Convexity and conditioning on Y give the outlet bound. For a smaller actual contrast the admissible curvature is larger, so the worst-contrast choice remains valid. □

Exact finite-jump proof. On one oriented edge let $u_s = k_{xy} w_s p_s(x)$ and $v_s = k_{yx} w_s p_s(y)$, before applying the species multiplier a_s. Interpolate $z_s(\theta) = v_s + \theta(u_s - v_s)$, $0 \leq \theta \leq 1$. The edge entropy production is exactly

$$\sigma_e = \int_0^1 \sum_s a_s \frac{z_s'(\theta)^2}{z_s(\theta)}\, d\theta,$$

by the integral formula for $\log u_s - \log v_s$. For two species set $Z = z_1 + z_2$, $r = z_1/Z$, $h = Z'/Z$ and $g = r'$. The edge contribution to $\dot{\mathcal{F}}_\psi$ is

$$-\int_0^1 Z\psi''(r)\{Qg^2 + (a-b)r(1-r)hg\}\, d\theta,$$

while the entropy integrand is $Z\{Ah^2 + 2(a-b)hg + [Q/(r(1-r))]g^2\}$. Apply the same positive-semidefinite matrix at each θ and sum all edges. Standard limits handle vanishing densities with finite entropy production. The multi-label information version uses the same interpolation and rank-two inequality. □

The curvature is locally maximal for a certificate of the form $\int m\psi(r)$: increasing it at any interior posterior makes the displayed matrix indefinite. This is a local statement, strengthened by the constructive operation below; local maximality alone would not prove global attainability.

4.1 Relation to the classical separation value

Let $V(r) = (2r-1)\log[r/(1-r)]$, so $V'' = 1/[r^2(1-r)^2]$. An exact decomposition is

$$\psi'' = \frac{4}{K}V'' + \frac{2}{r(1-r)[1+\sqrt{1+Kr(1-r)}]}.$$

Thus the leading weak-contrast term is the known Dirac value, with a specific finite-contrast correction. For normalized conditional densities,

$$\int mV(r) - V(f) = fD_{\mathrm{KL}}(p_1\|p_2) + (1-f)D_{\mathrm{KL}}(p_2\|p_1).$$

The correction's Jensen increase lies between $2\mathcal{I}/[1+\sqrt{1+K/4}]$ and $\mathcal{I}$. The endpoint law is therefore stronger than the mutual-information bound near accurate separation: the weighted symmetric relative entropy diverges as the output supports separate.

For target recovery R and product purity $P > f$, define

$$B = \frac{fR(1-P)}{(1-f)P}, \quad z = \frac{fR}{P}, \quad r_- = \frac{f(1-R)}{1-z}.$$

Here B is background acceptance, z the product fraction and r_- the target fraction in rejects. The complete-task bound is

$$\Sigma \geq z\psi(P) + (1-z)\psi(r_-). \tag{5}$$

It charges rejects and finite recovery. Reporting only the product purity would not specify this cost.

4.2 A consequential dilute-feed limit

The intended use was high-recovery purification of dilute valuable material. The endpoint theorem gives a sharp distinction from the reversible mixing-work benchmark. Keep $0 < R < 1$ and $0 < P < 1$ fixed while $f \downarrow 0$. Then

$$\boxed{\lim_{f\downarrow 0} \Sigma_{\mathrm{inf}}(f,R,P) = \frac{4}{K}\log\frac{1}{1-R}.} \tag{6}$$

This is entropy per input particle. The exact ideal infimum is earned by the network construction below. To prove the limit, use $z = fR/P \to 0$, $r_- = f(1-R)/(1-z)$ and $V(f) = \log(1/f) + o(1)$. Then

$$zV(P) + (1-z)V(r_-) - V(f) \longrightarrow \log\frac{1}{1-R}.$$

The finite-contrast correction is bounded by the retained mutual information, which tends to zero since it is at most the binary entropy of f. Thus the entropy per recovered target obeys

$$\frac{\Sigma_{\mathrm{inf}}}{fR} \sim \frac{4}{KfR}\log\frac{1}{1-R}.$$

In contrast, the reversible separation-work term satisfies $\mathcal{I}/(fR) = \log(1/f) + O(1)$. The full work in a restored apparatus is $k_BT(\Sigma + \mathcal{I})$ when the remaining marginal free energy is restored. Thus a device restricted to shared affinities faces a dilution penalty proportional to $1/f$ per recovered target, even when its thermodynamic control is ideal. Adding stages or operating more slowly does not change this leading penalty. Acquiring an energetic or accessibility difference between species changes the physical problem and can avoid this restriction.

For example, at $a = 1, b = 2, R = P = 0.99$, the limiting entropy per input is 36.84136149. Direct evaluation gives 36.84148268 at $f = 10^{-6}$ and 36.84136151 at $f = 10^{-10}$. These are model calculations, not material measurements. The limit is not uniform at $R = 1$ or $P = 1$, and does not apply unchanged when the requested purity decreases with the feed fraction.

5 An exact common-potential operation

Use one translating harmonic potential

$$\beta U(x,t) = \frac{c}{2}[x - z(t)]^2,$$

and start both species at its common equilibrium $N(0, 1/c)$. Their variances remain $1/c$ exactly, and their means satisfy

$$\dot{m}_a = -ac(m_a - z), \qquad \dot{m}_b = -bc(m_b - z).$$

The Gaussian law is earned by this specified linear diffusion; it is not assumed for an unknown receiver. Let r be the input prior and $d = m_a - m_b$. For $A = ra + (1-r)b$, $Q = (1-r)a + rb$, eliminating the shared center gives

$$\Sigma = \frac{1}{(b-a)^2}\int_0^T [A\dot{d}^2 + 2abc\, d\dot{d} + abQc^2d^2]dt.$$

With $d(0) = 0, d(T) = d_f$, strict convexity gives

$$d(t) = d_f \frac{\sinh(\gamma t)}{\sinh(\gamma T)}, \qquad \gamma = c\sqrt{abQ/A},$$

$$\Sigma_{\min}(T) = \frac{cd_f^2}{(b-a)^2}[ab + \sqrt{abAQ}\coth(\gamma T)].$$

An explicit shared waveform is obtained from

$$m_b(t) = -\frac{b}{b-a}\left[d(t) + ac\int_0^t d(s)ds\right], \qquad z(t) = m_b(t) - \frac{\dot{d}(t) + acd(t)}{c(b-a)}.$$

Its initial jump is included. A final common recentering to the weighted mean leaves joint free energy $r(1-r)cd_f^2/2$. The exact work ledger is

$$\beta W = \Sigma + \frac{1}{2}r(1-r)cd_f^2.$$

This residual is stored, not silently dissipated or recovered. Initial loading, finite collection barriers and actuator losses remain separate apparatus costs.

Writing $\epsilon = d_f\sqrt{c}$ and $K = (a-b)^2/(ab)$,

$$\Sigma_{\inf} = C(r)\epsilon^2, \qquad C(r) = \frac{1 + \sqrt{1 + Kr(1-r)}}{K}. \tag{7}$$

For equal priors and small ϵ, $\mathcal{I} = \epsilon^2/8 + O(\epsilon^4)$, and $\Sigma_{\inf} = \epsilon^2/(8\alpha)$. Hence the information coefficient is constructively sharp from an initially mixed equilibrium.

For a single threshold readout, any specified R, P in the interior is attained by the mean separation

$$\epsilon = \Phi^{-1}(R) - \Phi^{-1}(B),$$

where Φ is the standard normal CDF and B is from the previous section. This is a finite-error construction, including useful target recovery.

For a complete canonical two-outlet task, also reset the residual position within each outlet. The exact extra entropy is $\mathcal{I}(S;X) - \mathcal{I}(S;Y)$ by the common-mixture relaxation argument used below. Common marginal reshaping then restores auxiliary resources reversibly. The resulting complete trap comparison has

$$\Sigma_{\text{trap,complete}} = \Sigma_{\text{trap}} + \mathcal{I}(S;X) - \mathcal{I}(S;Y), \qquad \beta W_{\text{complete}} = \Sigma_{\text{trap}} + \mathcal{I}(S;X)$$

when the remaining species-blind free energy is restored. For equal-feed 99% purity/recovery, the extra entropy is 0.02864758; the infinite-time trap-plus-reset upper value is 89.24525005. This final reset is included in the complete-task table and figure.

6 Global construction without an external observer

A locally sharp cell is insufficient if routing and reset are assumed free. The next construction uses actual finite physical states for its history and only common energies and common gates. The proof is intentionally given first with a full history tree: its large resource requirement makes the mathematical infimum clear without hiding posterior compression in an observation oracle. A more compact score-grid composition is then tested numerically.

Attainability in the ideal common-affinity control class

Admissible model. There are two conserved labels with kinetic prefactors $a, b > 0$, $a \neq b$. Every protocol uses a finite set of physical states and a common, predetermined, piecewise smooth control schedule. On each enabled edge,

$$k_{ij}^{s}(t) = D_s \kappa_{ij}(t), \qquad D_1 = a, \quad D_0 = b, \qquad \frac{\kappa_{ij}(t)}{\kappa_{ji}(t)} = \exp\{E_i(t) - E_j(t)\}.$$

The dimensionless state energies E_i are common to both labels. Common gates may disable both directions of an edge. Arbitrarily many auxiliary states, arbitrarily long finite durations, and arbitrarily large finite energies are allowed as accuracy increases. Initial auxiliary registers contain no label information. Entropy production is the usual channel-resolved Markov entropy production, with each unordered edge counted once. Work performed by changing the common energies is included in the thermodynamic ledger but is not the quantity minimized here. No bound on apparatus size, duration, or actuator power is asserted.

Theorem (ideal entropy-production infimum). Fix an input target fraction $f \in (0, 1)$, desired product purity $P \in (f, 1)$, and desired recovery $\rho \in (0, 1)$. Set

$$k = \frac{(a-b)^2}{ab}, \qquad r_+ = P, \qquad q = \frac{f\rho}{P}, \qquad r_- = \frac{f(1-\rho)}{1 - f\rho/P}.$$

Let ψ be normalized by $\psi(f) = \psi'(f) = 0$ and satisfy

$$\psi''(r) = \frac{2\left[1 + \sqrt{1 + kr(1-r)}\right]}{kr^2(1-r)^2}.$$

In the admissible class, the infimum of entropy production over protocols starting unseparated and producing two exhaustive outlets with purity at least P and recovery at least ρ is

$$\boxed{\mathcal{B}(f, \rho, P) = q\psi(r_+) + (1-q)\psi(r_-).}$$

The infimum can be approached by finite common-affinity protocols whose auxiliary history registers are restored to blank states to arbitrary accuracy. Their graph size, energy range, and duration need not remain bounded. The lower bound is the convex-posterior entropy-production inequality proved separately. The construction below proves the matching upper bound.

Proof, step 1: an exact two-state weak cell. Use physical states $x = \pm 1$ and a common control $F(t)$, $|F(t)| < 1$, with rates

$$k_{-+}^{s} = \frac{D_s}{2}(1 + F), \qquad k_{+-}^{s} = \frac{D_s}{2}(1 - F).$$

These rates have the common affinity $\log(k^s_{-+}/k^s_{+-}) = 2\,\mathrm{atanh}\,F$; for example take $E_+ = -\,\mathrm{atanh}\,F$ and $E_- = +\,\mathrm{atanh}\,F$. If $\mu_s = \mathbb{E}_s x$, then, exactly,

$$\dot{\mu}_s = D_s(F - \mu_s), \qquad \sigma_s = D_s(F - \mu_s)[\mathrm{atanh}\,F - \mathrm{atanh}\,\mu_s]. \tag{A1}$$

Both species start with $\mu_s = 0$. At a node with posterior r, put

$$A = ra + (1-r)b, \quad Q = (1-r)a + rb, \quad \gamma = \sqrt{abQ/A}, \quad d(t) = \varepsilon\frac{\sinh(\gamma t)}{\sinh(\gamma T)}.$$

Writing $J(t) = \int_0^t d(u)\,du$, choose

$$\mu_a = -\frac{a}{b-a}(d + bJ), \quad \mu_b = -\frac{b}{b-a}(d + aJ), \quad F = -\frac{\dot{d} + (a+b)d + abJ}{b-a}. \tag{A2}$$

Then (A1) holds and $\mu_a(T) - \mu_b(T) = \varepsilon$. The quadratic part of the weighted entropy action is exactly

$$\mathcal{S}_2 = \frac{\varepsilon^2}{(b-a)^2}\left[ab + \sqrt{abAQ}\coth(\gamma T)\right]. \tag{A3}$$

Let $h = \varepsilon^2$ and take $T = \log(1/\varepsilon)/\min(a,b)$. On every fixed compact interval of posteriors in $(0,1)$, $\gamma \geq \min(a,b)$ and

$$|F(t)| + |\mu_a(t)| + |\mu_b(t)| \leq K\varepsilon e^{\gamma(t-T)}$$

for all sufficiently small ε, with a uniform finite K. Hence all rates remain strictly legal. Since

$$(x-y)\{\mathrm{atanh}\,x - \mathrm{atanh}\,y - (x-y)\} = O\Big((|x| + |y|)^2(x-y)^2\Big),$$

the difference between the exact action (A1) and (A3) is $O(h^2)$ after time integration. The finite-T excess in (A3) is also $O(h^2)$. Consequently

$$\mathcal{S}_{\mathrm{cell}}(r,\varepsilon) = C(r)h + O(h^2), \qquad C(r) = \frac{1 + \sqrt{1 + kr(1-r)}}{k}, \tag{A4}$$

uniformly on such a compact interval. No Gaussian approximation is used.

Step 2: the physical outcome supplies the posterior update. At the end of the cell, let $y = \pm 1$ denote its physical state, and write $\bar{\mu} = r\mu_a + (1-r)\mu_b$ and $t = r(1-r)$. The outcome probability and posterior are

$$p_y = \frac{1 + y\bar{\mu}}{2}, \qquad r_y = \frac{r(1 + y\mu_a)}{1 + y\bar{\mu}}.$$

In particular,

$$\mathbb{E}r_y = r, \quad \mathbb{E}(r_y - r)^2 = \frac{t^2\varepsilon^2}{1 - \bar{\mu}^2}, \quad \mathbb{E}(r_y - r)^3 = -\frac{2\bar{\mu}t^3\varepsilon^3}{(1-\bar{\mu}^2)^2}. \tag{A5}$$

The increments are bounded by $K_0\varepsilon$, and their fourth moments are $O(h^2)$. Taylor expansion on a slightly larger compact posterior interval therefore gives

$$\mathbb{E}\psi(r_y) - \psi(r) = C(r)h + O(h^2), \qquad \mathbb{E}V(r_y) - V(r) = \frac{h}{2} + O(h^2), \tag{A6}$$

where $V(r) = (2r-1)\log[r/(1-r)]$ and $V''(r) = [r(1-r)]^{-2}$. The local excess in (A4) relative to the potential increase is $O(h^2)$. All priors and controls used in the finite protocol are predetermined from the known input prior and the two rate constants; no label measurement is performed.

Step 3: retain a finite physical history tree. Each history is represented by its own physical node. After the two-state cell, its two occupied outcomes become the roots of its two children. Within a fixed child, the auxiliary state is initially definite for both labels. It can be split quasistatically into a common $(1/2, 1/2)$ auxiliary pair without changing the child posterior or producing entropy in the reversible limit. The next cell is then run with the control for that posterior. No history is erased at this stage. The implementation of these common splits with finite energies and time is justified in step 6.

Stop a branch when its posterior leaves $[r_-, r_+]$. Let τ be its stopping depth. Every visited or stopped posterior lies in a fixed compact subinterval of $(0, 1)$ for small ε, because the overshoot is $O(\varepsilon)$. From (A6), the expected increase of V per active cell is at least $h/4$. Boundedness of V on that compact interval implies, for all n,

$$\frac{h}{4}\mathbb{E}(\tau \wedge n) \le \mathbb{E}V(R_{\tau\wedge n}) - V(f) \le M.$$

Thus

$$\mathbb{E}\tau \le \frac{4M}{h}, \qquad \Pr(\tau > N) \le \frac{4M}{hN}. \tag{A7}$$

Truncate the history tree at $N = \lceil h^{-2} \rceil$ and designate its remaining active histories as reject. Its tail mass is $O(h)$. This tree has finitely many states for every h, although its size can grow exponentially in N.

Summing the cell excess in (A6) over the actual active population gives

$$\mathcal{S}_{\text{cells}} = \mathbb{E}\psi(R_{\tau\wedge N}) - \psi(f) + O(h^2\mathbb{E}\tau) = \mathbb{E}\psi(R_{\tau\wedge N}) + O(h). \tag{A8}$$

The upper and lower stopped posteriors have $O(\varepsilon)$ overshoot. Together with the $O(h)$ tail mass and conservation of the global prior, this makes the achieved outlet purity and recovery differ from the threshold targets by $O(\varepsilon)$.

Step 4: charge the final history erasure exactly. Let H denote the retained history and $Y \in \{-, +\}$ its assigned outlet. Inside one outlet, let q_j be the conditional history marginal, $r_j = \Pr(S = 1 \mid H = j)$, and $r_Y = \sum_j q_j r_j$. Choose a common energy $E_j = -\log q_j$ on the occupied history support and common heat-bath base rates $\kappa_{ij} = q_j$ for $i \neq j$. Each species then obeys

$$P_s(t) = q + e^{-D_s t}(P_s(0) - q).$$

Detailed balance gives the exact integrated relaxation entropy

$$\mathcal{S}_{\text{erase,Y}} = r_Y D_{\text{KL}}(P_1(0)\|q) + (1 - r_Y)D_{\text{KL}}(P_0(0)\|q) = \sum_j q_j D_{\text{KL}}[\text{Bern}(r_j)\|\,\text{Bern}(r_Y)]. \tag{A9}$$

After summing with outlet masses, this is $I(S; H \mid Y)$. Except for tail mass $O(h)$, the histories in each outlet have posterior width $O(\varepsilon)$. All posteriors remain in a fixed compact interval. The bounded curvature of binary negative entropy therefore gives

$$I(S; H \mid Y) = O(\varepsilon^2 + h) = O(h).$$

The corresponding loss of ψ has the same $O(h)$ bound. Once the histories have the common conditional distribution q, a common quasistatic compression can restore the history registers to a canonical blank state within each outlet with arbitrarily small additional entropy production. Its reversible work is not omitted: it cancels the corresponding marginal free-energy changes used to expand the registers earlier, apart from the prescribed input/output marginal change.

Consequently the complete ideal tree, including the irreversible part of final history erasure, satisfies

$$\mathcal{S}_{\text{ideal}} \le \sum_y \Pr(Y = y)\psi(\Pr(S = 1 \mid Y = y)) + O(h). \tag{A10}$$

Step 5: obtain the requested inequalities rather than just convergent endpoints. Choose the tree's internal purity and recovery targets to exceed P and ρ by $\sqrt{\varepsilon}$, for sufficiently small ε. Their thresholds remain in a common compact posterior interval. Their $O(\varepsilon)$ overshoot, truncation, and implementation errors are then dominated by the strict margins. The finite approximating protocol has purity at least P and recovery at least ρ, while its endpoint potential tends to $\mathcal{B}(f, \rho, P)$. Conversely, for any two-outlet protocol satisfying those inequalities, the high outlet posterior is at least r_+ and the low outlet posterior is at most r_-. At fixed mean f, its two-point posterior law is a mean-preserving spread of the desired endpoint law. Convexity of ψ makes the desired value a lower bound. This proves that the infimum for the inequality constraints is the stated value.

Step 6: finite implementations of the reversible operations. The preceding description uses reversible-limit common transfers, splits, and compressions. None is a label-aware primitive. The following elementary finite-state constructions justify their use in an infimum theorem.

Occupied-to-empty transfer. With an edge closed, place its empty destination higher by a common energy L. Opening and equilibrating the pair costs $\log(1 + e^{-L})$ per source particle when the destination is exactly empty. Both labels have the same initial conditional state and the same equilibrium. Slowly tilt the energy difference from $+L$ to $-L$, then close the edge. The final source leakage is $O(e^{-L})$, and entropy produced by the slow tilt tends to zero. A finite permutation can be decomposed into such transfers with an empty buffer. A split into an equiprobable pair uses the same procedure but ends at zero energy difference.

Common quasistatic reshaping. On any finite positive common distribution q, use base rates $\kappa_{ij}(t) = \pi_j(t)$ along a smooth positive path of common equilibrium distributions. Then $\dot{P}_s = D_s(\pi - P_s)$. Under a time dilation by L_t, the lag is $O(L_t^{-1})$ and the integrated entropy production is $O(L_t^{-1})$, with finite constants for the chosen positive path. Approximate zero probabilities by high but finite energies, and include the small initial/final relaxation costs. Final histories can thus be compressed to canonical outlet registers and all remaining auxiliaries returned arbitrarily close to blank states.

Finite error budget. For each fixed ε, the truncated tree has finitely many operations. If there are G common transfer/split operations, their leakage and entropy can be made smaller than any prescribed δ by choosing finite L and finite slow-operation times. For instance the cumulative leakage is bounded by a constant times Ge^{-L}, and the relaxation contribution from reusing slightly contaminated buffers is bounded by a finite-protocol constant times $(L + 1)e^{-L}$; choose L after G is fixed. Allocate at most δ/G additional slow-driving entropy to each operation. Heat-bath history relaxations converge exponentially, and common reshaping times can similarly be chosen to meet a total δ budget. Every resulting energy, rate, state count, and duration is finite.

Finite-state laws depend continuously on their initial laws and finite controls. Entropy production also has the needed finite-protocol continuity: use the common-energy identity "work minus nonequilibrium free-energy change", with $x \log x$ continuous at zero, rather than taking pointwise limits of logarithmic currents at empty states. Thus approximation errors can be chosen sufficiently small that the final outlet constraints and total entropy change by less than a prescribed tolerance. Choose these tolerances after the finite tree is fixed, then take the diagonal limit $\varepsilon \to 0$. The strict margins in step 5 absorb the final errors. This supplies genuine finite admissible protocols approaching (A10), and completes the proof. □

Corollary (retained-state binary classification). For equal label priors, normalize $\psi(1/2) = \psi'(1/2) = 0$. Its curvature is symmetric about $1/2$, hence $\psi(r) = \psi(1-r)$. If the Bayes classification error of the retained physical state is

$$e_{\mathrm{B}} = \mathbb{E}\min(R, 1 - R),$$

then convexity gives

$$\mathbb{E}\psi(R) = \mathbb{E}\psi(\min(R, 1-R)) \geq \psi(e_{\mathrm{B}}).$$

Since ψ decreases on $(0, 1/2)$, an error requirement $e_{\mathrm{B}} \leq e$ implies $\Sigma \geq \psi(e)$. The two symmetric outlet posteriors e and $1-e$, each of mass $1/2$, attain this ideal infimum by the theorem. Thus the exact ideal retained-state classification cost is $\psi(e)$. The decomposition

$$\psi''(r) = \frac{4}{kr^2(1-r)^2} + \frac{2}{r(1-r)[1+\sqrt{1+kr(1-r)}]}$$

shows that the non-Dirac correction to ψ has finite endpoint limits. Consequently, for fixed $k > 0$,

$$\boxed{\psi(e) = \frac{4}{k}\log\frac{1}{e} + O_k(1), \qquad e \downarrow 0.}$$

For weak relative kinetic contrast, $k \sim \delta^2$, giving the leading $4\delta^{-2}\log(1/e)$ cost. This concerns information retained in the modeled physical state, not an unmodeled observer's trajectory record.

Scope of the theorem. This is an ideal infimum over a union of finite controllable networks. It makes no fixed-memory, fixed-time, fixed-barrier, throughput, or manufacturability assertion. The modeled particle entropy production is approached with quasistatic common controls; a real actuator or fluidic implementation may add dissipation. Different species affinities, species selective state spaces, or an unmodeled trajectory observer are not covered. The construction does explicitly retain physical histories, charge their label-dependent erasure, and restore their label-independent memory content reversibly in the ideal limit. It does not assume a free label oracle.

6.1 Finite common operations and their actual resource cost

The elementary transfers were also tested at finite control ranges. Use a common two-well gap g, equilibrium probability $\pi = (1+e^g)^{-1}$, and rates $D_s\rho\pi$ and $D_s\rho(1-\pi)$. The nominally closed barrier has $\rho = e^{-B} > 0$, so leakage is represented. During a linear open-barrier sweep of duration T, the relative entropy $J = D_{\mathrm{KL}}(p\|\pi)$ obeys

$$\dot{J} = (p-\pi)\dot{g} - \sigma, \qquad \dot{p} = D_s(\pi - p).$$

Including equilibration after a sweep from $+L$ to $-L$ gives

$$\Sigma_{\mathrm{transfer}} \leq \log(1+e^{-L}) + \frac{2L}{T}\mathbb{E}[D_s^{-1}],$$

before the separately bounded finite-barrier ramp terms. Thus increasing T alone at fixed L leaves a preparation and leakage floor.

With $a = 1, b = 2$, equal weights, $L = 12$ and $B = 24$, the computed transfer entropy decreases from 1.78891885 at sweep duration 10 to 0.00180612 at duration 10,000. Each operation additionally includes finite setup, restoration and equilibration totaling 42 time units. The final incorrect-well probability is about 6.1442×10^{-6}. A common binary reset at the long duration requires work $0.69351730\, k_B T$, consistent with $W/k_BT = \Sigma + \log 2 - H_b(p_{\mathrm{final}})$; it is not a free operation. Independent current-affinity integration and the work balance agree within 9×10^{-12}. These are numerical controls of the finite-state model. They support the finite approximation lemma while exposing why the full composition's reversible routing limit does not establish useful throughput.

6.2 An exact auxiliary Gaussian-channel identity

The earlier harmonic route remains useful. With $h = \epsilon^2$, its posterior has the same law as the diffusion $dr_s = r_s(1 - r_s)dW_s$ run for time h. This is a statistical identity for the produced spatial distribution; it does not supply a free external measurement. Since $\psi''(r)r^2(1 - r)^2/2 = C(r)$, Ito's formula gives

$$\mathbb{E}[\psi(r') - \psi(r)] = \int_0^h \mathbb{E}C(r_s)\, ds.$$

Concavity of C and $|C''| \le 1 + K/4$ yield the exact finite-strength estimate

$$0 \le C(r)h - \mathbb{E}\Delta\psi \le \frac{1 + K/4}{64}h^2. \tag{8}$$

The Dirac increment is exactly $\mathbb{E}\Delta V = h/2$. These identities support the local harmonic construction and remain in the executable evidence. The global finite-network theorem above uses exact binary cells and therefore needs no unproved Gaussian lattice approximation.

7 Executable evidence and exact scope

Independent checks reconstructed the quadratic and finite-edge proofs. The saved tests include 20,000 continuum cases, 20,000 information-flow edges and 200 finite-jump potential edges. No tested inequality was violated; endpoint and interpolated edge calculations agreed to about 10^{-13}. These finite checks support implementation and algebra; the proofs establish universality.

The common harmonic operation was separately integrated as an ODE. At $a = 1, b = 2, c = 1$, $d_f = 0.001$ and $\gamma T = 7$,

$$\frac{\mathcal{I}}{\alpha\Sigma} = 0.999999018.$$

For a 99%/99% equal-feed task with $\gamma T = 3$, the entropy is 89.44482394, work 92.15077116 and retained joint free energy 2.70594722. The independently computed ledger residual is below 7×10^{-14}. The optimal same-duration trap dissipates less than the constant-speed trap (111.6002), while no superiority over every existing separator is claimed.

Table 1: Complete two-outlet tasks in the declared model. Entries are dimensionless entropy per input particle; physical entropy is k_B times the entry. The single-trap column is an explicit upper construction within its control family, not a claim of optimality among all devices.

b/a	Feed f	R	P	Endpoint floor	Trap + reset infimum
2.00	0.50	0.99	0.99	36.6486	89.2453
1.10	0.50	0.99	0.99	1982.0526	4765.2002
2.00	0.10	0.95	0.95	25.2614	70.8275

Table 2: Deterministic composition of exact two-state cells, $a = 1, b = 2, f = 0.1$, with requested $R = P = 0.95$. The entropy includes actual cell dynamics and conditional-information reset; common routing and other common resets are in their reversible limits. The floor is evaluated at the achieved outlets, including overshoot.

ϵ	Actual P	Actual R	Total Σ	Achieved floor	Mean cells
0.100	0.952514	0.951724	26.256894	25.582069	568.7
0.075	0.951796	0.951317	25.800403	25.502379	1057.4
0.050	0.951132	0.950944	25.557848	25.429821	2453.6
0.025	0.950552	0.950511	25.379229	25.350789	9969.6

The requested endpoint floor is 25.26142485. At the finest recorded strength, the achieved outlet floor is 25.35078881, and the total modeled entropy is 25.37922929, a 0.1122% excess. This calculation retains 2,058 score bins and uses 185,112 clock cycles to reduce unfinished mass below 10^{-10}. The mean number of active cells per input is 9,969.6. Species mass balances close to 1.5×10^{-14}. The entropy-cost interpolant was checked against independent quadrature to absolute error below 2.3×10^{-15} per cell. These are computational model results; finite routing times, leakage and actuator dissipation are not part of the simulation.

Across 36 weighted-prior module checks, the posterior martingale residual was below 1.2×10^{-16} and the work-ledger residual below 4.6×10^{-14}. At $\epsilon = 0.001$, direct posterior quadrature matches $C(r)\epsilon^2$ to relative error below 3.6×10^{-9} for the tested contrasts and priors. The exact finite-strength bound (8) explains this agreement.

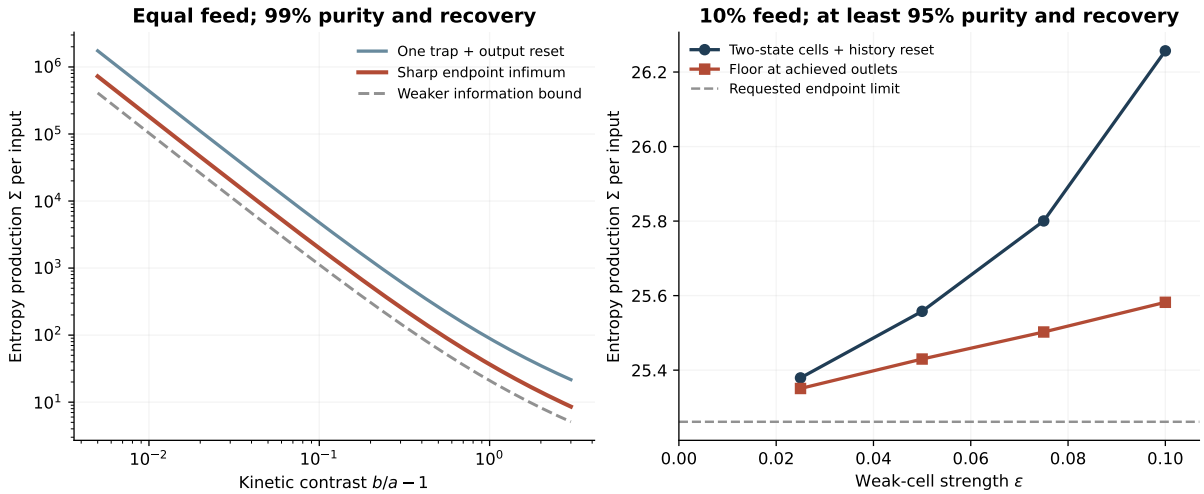


Figure 1: Left: ideal common-affinity entropy bounds and the complete single harmonic operation. Right: deterministic composition of exact two-state cells approaches the endpoint bound as cell strength decreases. Actual cell entropy and discarded history are charged; common routing and auxiliary resets are in the reversible limit. No material or device measurement appears in this figure.

8 Strongest comparisons and actual novelty boundary

Kedem–Caplan's degree-of-coupling efficiency law, and its far-from-equilibrium stationary extension, give exactly the local square-root coefficient after mapping the two thermodynamic forces to common motion and composition motion [3, 4]. Picone et al. derive the Dirac value from an Onsager transport framework, using small element separation and a specified separative power definition [5]. The present result connects these ingredients to actual integrated entropy production, finite contrast and arbitrary nonstationary networks. A search failing to find that exact connection is not evidence of historical priority.

The nearest retained-output comparison is Hasegawa's October 2026 preprint *Thermodynamics of Classifiers* [8]. It considers initially identical states evolving under class-dependent, time-independent Markov generators. Its entropy bound on terminal classification error involves the history functional $\int_0^T \sqrt{\Sigma_0(t) + \Sigma_1(t)}\, dt/t$. The present result imposes common affinities and finite kinetic contrast, permits shared time-dependent control, and obtains an explicit attainable endpoint cost. General thermodynamic bounds for retained classification are therefore prior art.

Sartori and Pigolotti explicitly distinguish kinetic from energetic copying [9]. Setting their energetic discrimination parameter to zero makes the forward and reverse right/wrong rate ratios equal.

Their transient single-bit calculation uses a three-state construction, while their cost optimization studies specified copying and proofreading networks. Their mechanism is relevant prior art; no claim is made to invent discrimination through kinetic barriers.

Kolchinsky and Wolpert already give broad constrained-protocol entropy bounds and reversible constructions [10]. Their reversible correlation construction permits independently setting selected transition activities to zero. Applying that freedom separately to the two label-conditioned copies supplies unbounded kinetic selectivity, which the present finite-contrast hypothesis excludes. Thus shared energies alone do not establish the new bound. The general methodology of integrating a local dissipation inequality is also established; the contribution here is the explicit optimal value and matching constrained operation.

Modern shared-potential ratchets already engineer diffusivity-dependent transport and optimize driving costs [6]. The inspected open preprint also simulates separated spatial clouds; it is not dismissed as a mean-drift proposal. Its analytic cost is periodic-state bath heat for a prescribed loop, not the present fixed purity/recovery endpoint entropy. Drift per cycle alone does not specify packet dispersion, collected recovery or the complete process budget. The final published full text was not available in the inspected route, so detailed preprint formulas are not attributed automatically to that version.

Chen et al. derive a thermodynamic-geodesic finite-time excess-work bound for complete membrane separation [7]. Their governing model uses two perfectly selective membranes, each admitting only one species. Those are stronger physical resources than the finite-contrast common-affinity class. Their quasistatic limit therefore does not contradict a positive kinetic-only entropy floor.

Current Astra N1079 was read through its governing derivations and ordinary countercurrent construction. That source correctly shows that stored intermediate pools escape a one-source turnover obstruction. The present theorem permits such pools and does not infer release kinetics from bulk diffusivity. The exact pinned source paths, hashes and use boundary are retained in `astra_and_preservation.json`.

9 Failures that must remain visible

Trajectory information is different. At equilibrium, two species can have the same position distribution and zero particle entropy production while different jump counts reveal their rates to an external observer. Ideal Brownian quadratic variation can even reveal diffusivity. The theorem bounds stored physical label–state correlations. A recording apparatus is covered only after its states, affinities and costs are included. It is not a universal cost of knowledge or external observation.

Bounded rate ratios alone are insufficient. If the two directions of an edge acquire species-dependent affinities, arbitrarily slow common-looking protocols can create correlations through energetic discrimination. The common affinity condition is essential. Selective binding, different charges and species-specific pathways can supply precisely the resource the theorem excludes.

A restricted wash is not all separation. Uniform passive-chamber retention obeys a positive spectral-mixture inequality $Y(\rho D, t) \geq Y(D, t)^{\rho}$. It constrains a one-way retention task, but ordinary stored-pool counterflow can outperform that restricted operation. Both survive as scoped results, without turning either into a universal impossibility.

An ideal thermodynamic infimum is not throughput. The weak-operation construction has diverging numbers of operations and increasing resolution as it approaches the bound. Finite-width gates, leakage, interaction forces, controller erasure, finite density, solvent power and calibration can dominate an actual apparatus. The simple shared-trap protocol is fully solved within its model; the elaborate limiting network is not a fabricated machine.

Reproducibility and preservation

The entry point is `README.txt`, which gives commands for every main computation. The deterministic retained-score composition, independent inequality audit, exact two-state cells, harmonic controls, dilute-feed limit, finite common operations and failed candidate mechanisms are included with their recorded outputs. Python, NumPy, SciPy and Matplotlib reproduce the computations and figures. The LaTeX source and included constructive-theorem file reproduce this report.

The included preceding archive is byte-identical to the completed acquisition research archive, and recursively preserves the earlier port and excursion work. The new programme does not use those theorems as premises.

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